Cells and Gauge

File: gauge.kicad_sch

3V3 Rail

File: power.kicad_sch

MCU

File: mcu.kicad_sch

USB Port 1

IRQD
IRQ_N
EN_5V
EN_5V1
ADCIN1
P1_LDO_3V3
ADCIN2
LDO_3V3D
P1_LDO_3V3

File: usb_port.kicad_sch

USB Port 2

IRQD
IRQ_N
EN_5V
EN_5V2
ADCIN1
GND
ADCIN2
LDO_3V3D
GND

File: usb_port.kicad_sch

TPS25751 ADDRESS / DEAD-BATTERY STRAPS (SafeMode)
Port 1: ADCIN1 tied to its LDO_3V3, ADCIN2 to GND -> I2C 0x20
Port 2: ADCIN1 and ADCIN2 to GND   -> I2C 0x21
Same SDA/SCL bus (CC2340 I2C0, 100 kHz): gauge BQ40Z50-R2 0x0B

MONOLITH PRODUCTION BOARD - SHEET INDEX
2. Cells and Gauge: welded 4S pack pads, BQ40Z50-R2 gauge, protection FETs, BQ294700 + F102 second-level protection, SMBus pogo pads
3. 3V3 Rail: TPS62177 always-on 3V3
4. MCU: CC2340R5MODA, Tag-Connect TC2030-NL SWD pads
5. USB Port 1 / 6. USB Port 2 (one sheet file): TPS25751D + TPD2S300 + BQ25756 + TPS62933 per port
Rails (power symbols): PACK_P, 3V3, GND, CELL_N (gauge VSS only)
Spec and open items: Production/README.md. Derived from the dev board rev A8 (DevBoard/, history in DevBoard/README.md).

REVISION LOG
P0 2026-10-17 Dev board rev A8 with the dev-only parts removed (Production/tools/make_production.py, tidy_production.py):
                sleep-current section, bench supply, cell holders, headers, buttons, status LED, test points, DNP options
                and OR links. Added pack wire pads, gauge pogo pads, TC2030-NL SWD pads, factory test pads.
                Author: Claude Code. Reviewer: none yet.
Dev board history (A0-A8): DevBoard/README.md and the DevBoard root sheet.

P0: dev board rev A8 stripped by Production/tools/make_production.py; not reviewed
Spec + open items: Production/README.md
Sheet index, PD address straps
Monolith 100 W battery bank — production

Sheet: /
File: Monolith.kicad_sch

Title: Monolith production board — top level

Size: A3	Date: 2026-10-17	**Rev: P0**
KiCad E.D.A. 10.0.7		Id: 1/6

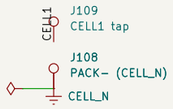
PACK WIRES (welded 4S EVE INR21700-58E)

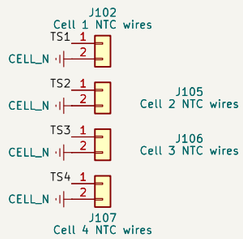
J108 PACK- and J112 PACK+: 16 AWG leads in 1.7 mm plated holes. J109-J111: cell tap wires.
Connect with USB unplugged, in order: PACK-, CELL1, CELL2, CELL3, PACK+ (note 15).



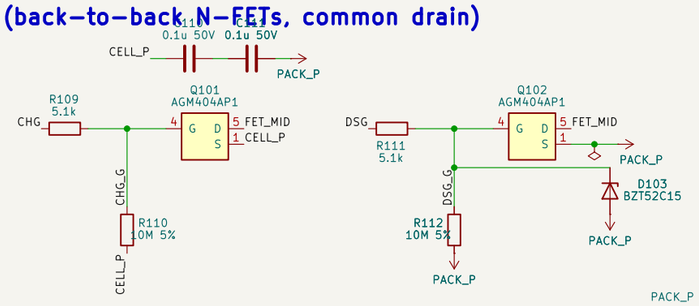


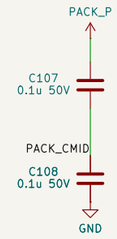




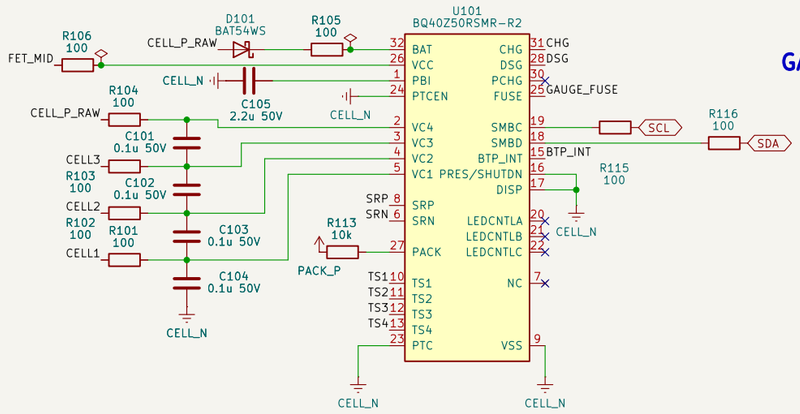


PROTECTION FETs (back-to-back N-FETs, common drain)

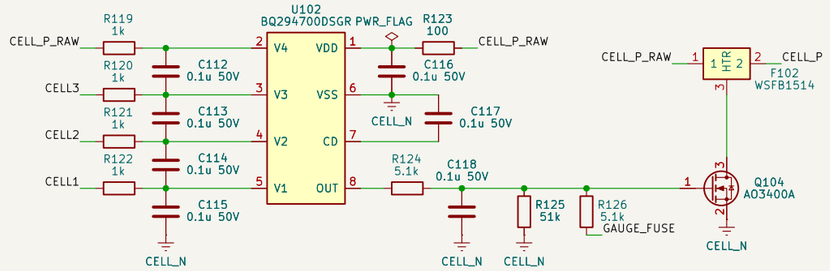




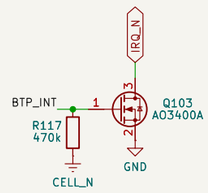
GAS GAUGE BQ40Z50-R2 (TI fig. 9-1)



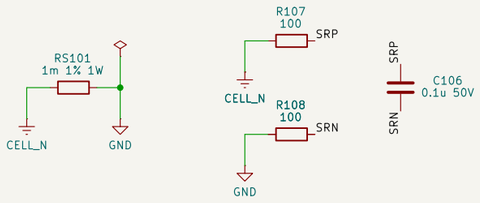
SECOND-LEVEL PROTECTION BQ294700 + SCP (TI fig. 9-11)



GAUGE SMBus (0x0B) + POGO PADS



LOW-SIDE CURRENT SENSE



FACTORY TEST PADS



Training: J113/J114 = learning-cycle load wires (20 AWG). WAKE: bridge TP113 to TP106 (PACK_P).

NOTES
1. CELL_N = gauge VSS (cell 1 negative). GND = system ground, on the far side of RS101.
2. Kelvin-route SRP/SRN from the RS101 pads: R107/R108/C106 at the IC. RS101 is a 1206 1 W 75 ppm alloy part (P0).
3. VC taps are separate thin traces from the tap pads J109-J111 and from the J108/J112 pads:
   VC4 from CELL_P_RAW (before F102) so fuse drop is not read as cell voltage.
4. TS1-TS4: one wired 10k NTC per cell on J102, J105-J107, taped to the cell at mid-height; insulated sensors only
   (cans 0-12.6 V above VSS; TS 2.1 V max). PTC, PTCEN, PRES, DISP tied to VSS per datasheet.
5. VCC fed from FET common drain via R106 (TI R13). PACK sense via R113 10k (TI R15).
6. BAT fed from CELL_P_RAW (cell top, before F102) via D101 + R105 (TI fig. 9-12).
7. Diamonds are PWR_FLAGs (ERC only, no part): each sits where its rail is sourced through a passive part.
8. Q103 buffers BTP_INT onto the shared open-drain IRQ_N (CC2340 DIO11): IRQ_N is low while BTP_INT is high.
   Keep BTP disabled, or set BTP_POL = 1 with BTP_EN = 1, or IRQ_N is held low (review 4 D-2).
9. D103 clamps Q102 VGS at 15 V in a hard short. The dev board's C109/D102 inrush options are not on P0.
10. No wake button: the gauge leaves SHUTDOWN when PACK_P comes up (USB plugged in: TPS25751 dead-battery sink +
    BQ25756 charging with 3V3 down, USB sheet note 11). P0 fallback: bridge TP113 (WAKE, R118 4.7k from CELL_P) to TP106.
11. Gauge access is the GAUGE Tag-Connect J302 on the MCU sheet (SMBus + CHG_INH + NRST): BQStudio / EV2400 for
    training, golden image and calibration. J113/J114 = 20 AWG load wires (PACK_P/GND) for the learning-cycle discharge.
12. TP106 PACK_P and TP109 GND are 1.0 mm probe pads.
13. ESD caps (TI): C110+C111 in series across the FET pair (fig. 9-2 C1/C2). C107+C108 across PACK+/PACK- (sec. 9.2.2.1.5).
14. Second level (TI fig. 9-11): U102 BQ294700 (4.35 V/cell, CD 0.1u = 1-2 s) or the gauge FUSE pin turns Q104 on;
    F102's heater then blows its fuse, permanently opening the cell top. F102 is also the pack's only overcurrent fuse.
15. Connect the pack with USB unplugged, PACK- first, then the taps bottom-up, then PACK+: a floating tap can read
    > 4.35 V, U102 trips after 1-2 s and F102 blows. Gauge data flash: COV Delay 1 s, Min Blow Fuse Voltage 10500 mV.

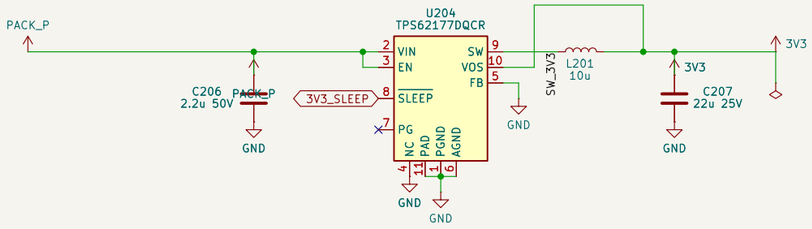
P0: dev board rev A8 stripped by Production/tools/make_production.py; not reviewed
Spec + open items: Production/README.md
Welded 4S pack pads, BQ40Z50-R2 gauge/protector, protection FETs
Monolith 100 W battery bank — production

Sheet: /Cells and Gauge/
File: gauge.kicad_sch

Title: Monolith production board — cells and gauge

Size: A3	Date: 2026-10-17	**Rev: P0**
KiCad E.D.A. 10.0.7		Id: 2/6

3.3 V ALWAYS-ON RAIL (TPS62177, 1 MHz, sleep mode under MCU control)



PACK_P SHARED BULK (both chargers + both 5 V buck inputs)



NOTES
1. PACK_P -> TPS62177 -> 3V3 (MCU + both TPS25751). Gauge BAT is NOT on this rail (TI: gauge must stay powered with the FETs off).
2. SLEEP low = sleep mode (internal 1.4 uA pull-down); CC2340 DIO21 (3V3_SLEEP) drives it, and R313 220k holds it high while U301 is in reset or unprogrammed.
3. PG is not used.
4. U204 pin 4 (NC) goes to GND, as TI recommends.
5. Diamonds are PWR_FLAGs (ERC only, no part): each sits where its rail is sourced through a passive part.
6. The dev board's sleep-current section (R201 R_MEAS, 2 x INA190, HT7550-1) is not on P0: PACK_LP is PACK_P. Measure sleep current on P0 with a meter in series with one pack lead (J108 or J112).
7. C208 = the one battery-side bulk cap for both chargers and both 5 V buck inputs (was C432/C532, one per port). Place it between the two power stages on PACK_P, next to both RBAT pads (USB sheet note 5).

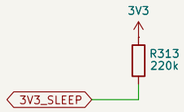
P0: dev board rev A8 stripped by Production/tools/make_production.py; not reviewed
Spec + open items: Production/README.md
TPS62177 3V3 always-on rail
Monolith 100 W battery bank — production

Sheet: /3V3 Rail/
File: power.kicad_sch

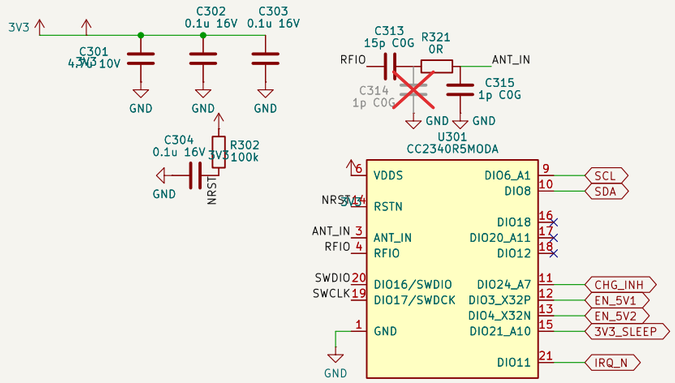
Title: Monolith production board — 3V3 rail

Size: A3	Date: 2026-10-17	**Rev: P0**
KiCad E.D.A. 10.0.7		Id: 3/6

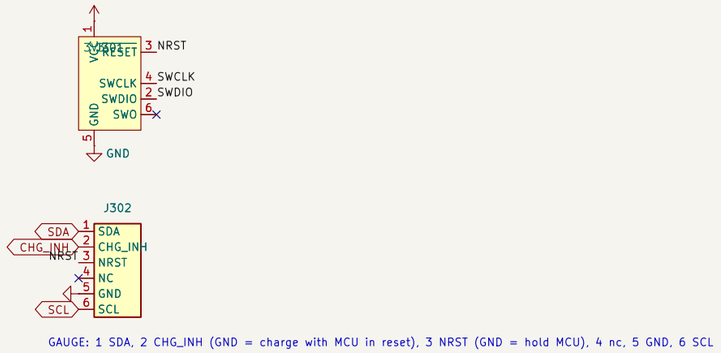
3V3_SLEEP PULL-UP



BLUETOOTH MCU CC2340R5MODA (per TI SWRS349A fig. 9-1)

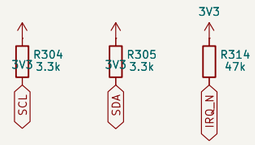


SWD (Tag-Connect TC2030-NL)



GAUGE: 1 SDA, 2 CHG_INH (GND = charge with MCU in reset), 3 NRST (GND = hold MCU), 4 nc, 5 GND, 6 SCL

I2C0 + IRQ PULL-UPS



NOTES
1. U301 antenna end at a board edge, ground 10-20 mm each side; keep-out in the footprint (fig. 9-2).
2. C301-C303 at the VDDS pads. R302 + C304 = power-on reset (SWRS349A fig. 9-1 R1/C92); no reset button.
3. C313 = CA1. R321/C314/C315 = Z61/Z60/Z62 pi match (TI: 0R, DNP, 1 pF); retune on the board.
4. No 32 kHz crystal: LFOSC calibrated against the 48 MHz crystal. DIO3/DIO4 are GPIOs.
5. Pins: DIO6 SCL, DIO8 SDA, DIO24 CHG_INH, DIO3/4 EN_5V1/2, DIO21 3V3_SLEEP, DIO11 IRQ_N.
   Spare (no-connect): DIO12, DIO18, DIO20, the dev board's LED and UART pins (docs/devboard/cc2340-pin-budget.md).
6. IRQ_N = TPS25751 IRQ x2 + gauge BTP (Q103), all open-drain. One pull-up: R314 47k.
   Firmware polls each source to find which.
7. Diamonds are PWR_FLAGs (ERC only, no part) where a rail is sourced through a passive part.
8. J301 = Tag-Connect TC2030-NL pads (no part fitted): XDS110 through a TC2030-IDC-NL cable. J302 (GAUGE) uses the same cab
   SWO (pin 6) not used. The debugger drives NRST. Bootloader: ROM bootloader on a blank part or the CCFG backdoor pin.
9. R313 220k holds 3V3_SLEEP high in reset (pins float; TPS62177 pulls down 1.4 uA typ).
   Costs 15 uA while driven low in S2. DIO21 is also the default ROM-bootloader pin:
   disable or move that trigger in the CCFG before the first flash (review 4 A-1).
10. R304/R305 SMBus pull-ups go to 3V3. Gauge work on J302 only with 3V3 up and U301 off the bus: hold NRST low (J302 pin 3)
    and ground CHG_INH (J302 pin 2) to keep charging enabled; an unpowered CC2340 can't take bus voltage (VDDS + 0.3 V).
11. DIO24 = CHG_INH: LOW allows charging. High or hi-Z stops both chargers
    (R417/R517 pull CE up), so reset/hang/unprogrammed stops charging.
12. Watchdog: CC2340 WDT runs in active, idle and standby and can't be stopped once enabled.
    GPIOs latch in standby; IRQ_N wakes it on an edge (SWRS349A 8.6, 8.11).
13. Temps: gauge DAStatus2 (cells, FETs); charger NTCs via TPS25751 'I2Cr' (BQ25756 REG0x37).
14. TP301 3V3 is a 1.0 mm probe pad.

FACTORY TEST PAD



P0: dev board rev A8 stripped by Production/tools/make_production.py; not reviewed
Spec + open items: Production/README.md
CC2340R5MODA, TC2030-NL SWD pads
Monolith 100 W battery bank — production

Sheet: /MCU/
File: mcu.kicad_sch

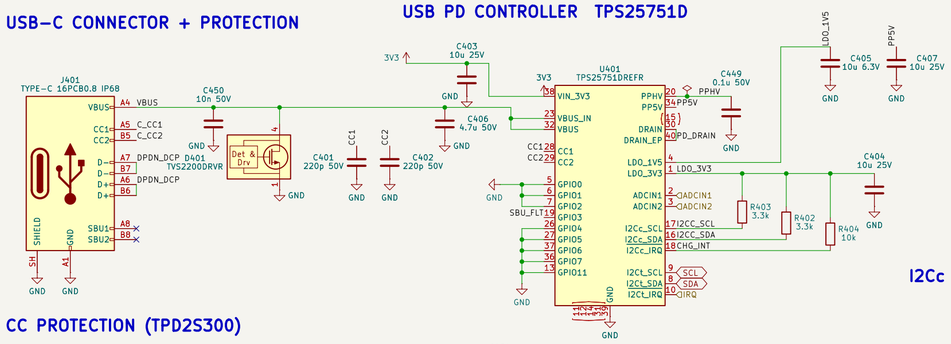
Title: Monolith production board — MCU

Size: A3	Date: 2026-10-17	**Rev: P0**
KiCad E.D.A. 10.0.7		Id: 4/6

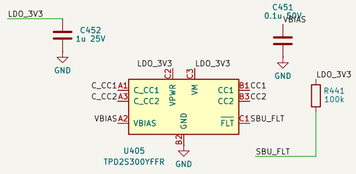
Charger battery side joins PACK_P here (net tie R430 removed).
Route SRN (R429) and the FB divider top (R418) from R427's PACK_P pad,
not from the pour: DRC can't catch a tap on the bus.

USB-C CONNECTOR + PROTECTION

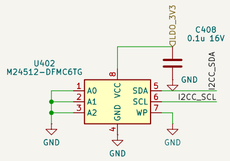
USB PD CONTROLLER TPS25751D



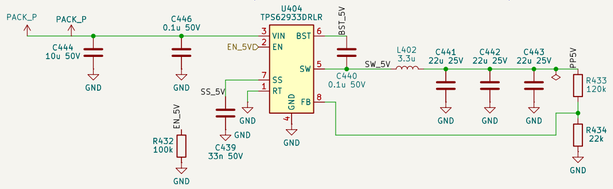
CC PROTECTION (TPD2S300)



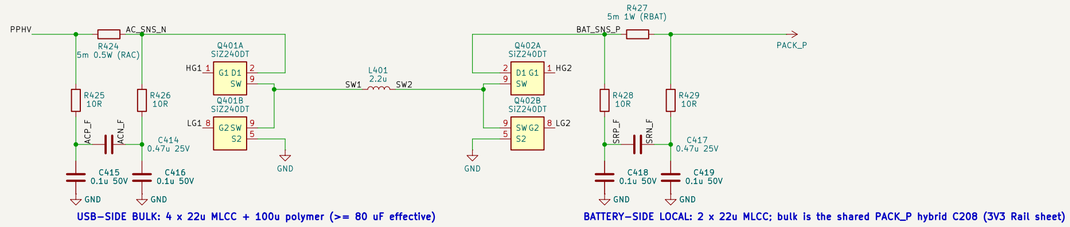
I2Cc BUS: EEPROM + PULL-UPS



PP5V SUPPLY TPS62933 (1.2 MHz, 5.16 V, 3 x 22u out) — 5 V source + VCONN

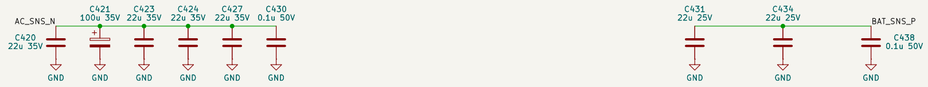


BQ25756 4-SWITCH BUCK-BOOST POWER STAGE (copy production layout exactly)

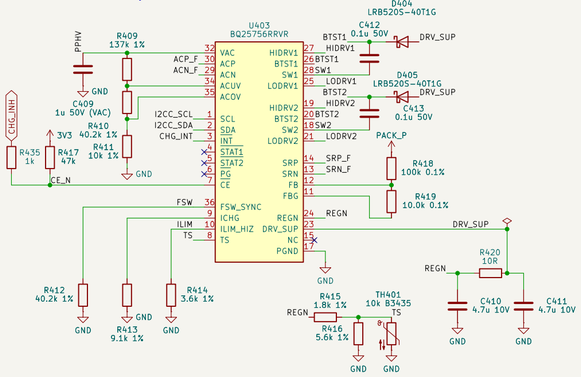


USB-SIDE BULK: 4 x 22u MLCC + 100u polymer (>= 80 uF effective)

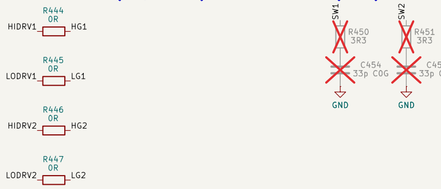
BATTERY-SIDE LOCAL: 2 x 22u MLCC; bulk is the shared PACK_P hybrid C208 (3V3 Rail sheet)



BQ25756 CONTROLLER (600 kHz, ICHG 5.5 A max, ILIM 5.6 A, VAC 4.1-22.5 V window)



GATE LINKS (0R, fitted) + SNUBBERS (DNP): P0 tuning hedge



PROBE PADS (1.0 mm): VBUS PPHV PP5V REGN SW1 SW2 + GND



NOTES (this sheet is used twice: USB Port 1 = 4xx, USB Port 2 = 5xx)
1. Address straps are on the root sheet: port 1 ADCIN1 = LDO_3V3, ADCIN2 = GND (0x20);
   port 2 ADCIN1 = ADCIN2 = GND (0x21). Both are SafeMode (TPS25751 table 8-6).
2. Unused TPS25751 GPIOs (0, 1, 2, 4-7, 11) tied to GND per datasheet (KiCad ERC flags this; expected).
   Never configure them as outputs.
3. CC is protected by U405 (TPD2S300; CC short-to-VBUS OVP, 24 V DC, + IEC 61000-4-2 ESD; the TPS25751 CC pins are
   26 V abs max but only +-1 kV HBM). VPWR/VM on LDO_3V3 (normal mode). VBIAS C451 0.1u 50V (>= 35 V needed).
   C401/C402 (220 pF) on the controller side: with the TPD2S300's 30-120 pF the total stays under 480 pF.
   D+ and D- are tied together at the connector (net DPDN_DCP: passive BC1.2 DCP); the TPS25751 does no BC1.2.
   No liquid detection and no SBU use on production (IP68 connectors, potted board): the connector SBU pins are
   no-connects; GPIO4-7 are tied to GND (note 2). U405 FLT -> GPIO5 = Fault_Input_Event (SBU_FLT, R441);
   a CC short to VBUS forces error recovery.
   D401 = TVS2200 flat clamp on VBUS (28 V at 40 A 8/20 us; TI's pick for the TPS25751).
4. Kelvin-route ACP/ACN from R424 and SRP/SRN from R427: filter parts at the IC.
5. Two MLCCs of each bulk bank plus C430/C438 at the FETs: the USB-side polymer C421 and the remaining MLCCs
   one row back. Polymer carries the bulk (no DC-bias loss), MLCCs carry the switching-frequency ripple.
   Battery side: 2 x 22u per port at the FETs: both ports, both TPS62933 inputs (1 x 10u 50V each) and the one shared
   PACK_P hybrid C208 form one bus: about 99 uF effective worst case, >= the BQ25756 80 uF C_OUT minimum each charger
   sees through its RBAT (Production/tools/pack_bus_sims.py A2). Still on hold: C427 (USB side; the USB sides are
   separate per port, so they can't share bulk).
6. I2Cc pull-ups go to this TPS25751's own LDO_3V3 (TI guidance).
7. The charger battery side is PACK_P directly; take SRN/FB from R427's pad, not from the pour.
8. VAC is taken before R424 per TI typical application; DRV_SUP fed from REGN via R420.
9. PPHV bulk (BQ25756 input bank) exceeds the 100 uF USB PD cSnkBulkPd: the TPS25751D sink switch soft-start limits
   inrush (C x SS: about 125 uF effective x 0.47 V/ms max = 0.06 A at SS setting 0).
   Set PPHV soft-start to setting 0 or 1 in the TPS25751 configuration.
10. Diamonds are PWR_FLAGs (ERC only, no part): each sits where its rail is sourced through a passive part.
11. CHG_INH: R417 pulls CE up to 3V3, so charging needs the MCU (DIO24) to drive CHG_INH low through R435.
    Reset, WDT reset, the ROM bootloader or an unprogrammed MCU stop charging: CC2340 Standby keeps the pin state.
    A dead pack (3V3 down) still charges. Keep DIS_CE_PIN = 0.
12. Gate drive: R444-R447 are 0R 0402 links at the gates (P0 hedge, no dev board): swap to 2.2-10R if SW1/SW2 ring;
    R450/C454, R451/C455 are DNP 0402 RC snubbers (3R3 + 33p). No turn-off diodes; BTST straight to its cap. Keep the
    gate traces short and wide (TI: >= 0.5 mm) with the FETs right at the BQ25756.
    Each leg is one Vishay SiZ240DT half-bridge (Q401/Q402: unit A = high side, unit B = low side, pin 9 = SW).
    40 V, 12.25/13.30 mOhm max at 4.5 V, Qg 6.9/6.5 nC. Worst case (20 V 5 A out from a 10 V pack) its fully-on
    high side dissipates ~1.8 W in one 3.3 x 3.3 package; needs the paddles' via arrays and the gap pad; the
    thermal throttle covers it. Buy from DigiKey/Mouser (JLC stock is near zero).
13. TP401 VBUS is a 1.0 mm probe pad. STAT1/STAT2/PG (U403 pins 4-6) are open-drain and unused (no pull-ups):
    status comes over I2C (TPS25751 I2Cr); set DIS_STAT_PINS = 1 in the BQ25756 configuration.
14. PP5V output: C441-C443 3 x 22u 1206 (about 33 uF effective) plus C407 10u at the PP5V pin. Each port's PP5V is
    its own TPS62933, a dedicated supply: USB PD cSrcBulk (10 uF min) applies, not the 120 uF cSrcBulkShared the
    TPS25751 table cites. Estimated 0.5 -> 3 A step undershoot about 190 mV (cap sims P3); measure it on P0.
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    Worst case (20 V 5 A reverse from a 10 V pack): ~10.7 A rms, ~1.6 W DCR loss, L near 1.7 uH at the 12 A peak
    (L/RSNS 0.34 uH/mOhm, inside TI's 0.05-1.26). Check its temperature on P0: the thermal throttle covers it.
16. RAC R424 is 0805 0.5 W (<= 0.16 W at 5.6 A). RBAT R427 1206 1 W (~0.5 W at 10 A). Both stay 5 mOhm, so ICHG/ILIM
    settings are unchanged. R420 is 0402: 0201 is used only where both nets stay <= 5.5 V.

PO: dev board rev A8 stripped by Production/tools/make_production.py; not reviewed
Spec + open items: Production/README.md
TPS25751D + BQ25756 + TPS62933; port 1 = 4xx, port 2 = 5xx
Monolith 100 W battery bank — production

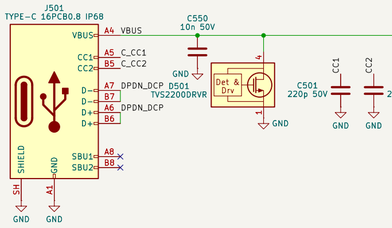
Sheet: /USB Port 1/
File: usb_port.kicad_sch

Title: Monolith production board — USB-C port

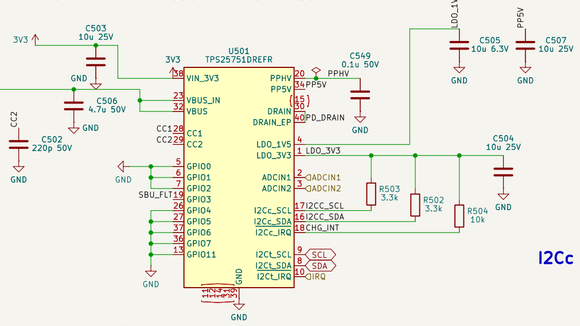
Size: A2	Date: 2026-10-17	**Rev: P0**
KiCad E.D.A. 10.0.7		Id: 5/6

Charger battery side joins PACK_P here (net tie R430 removed).
Route SRN (R429) and the FB divider top (R418) from R427's PACK_P pad,
not from the pour: DRC can't catch a tap on the bus.

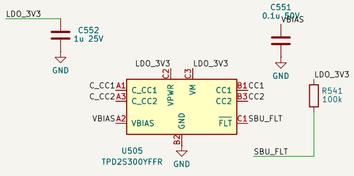
USB-C CONNECTOR + PROTECTION



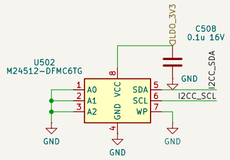
USB PD CONTROLLER TPS25751D



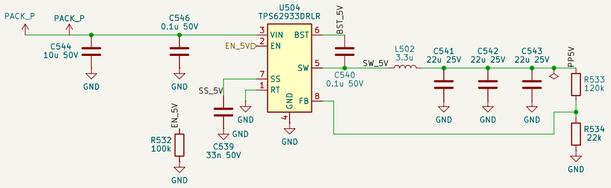
CC PROTECTION (TPD2S300)



I2Cc BUS: EEPROM + PULL-UPS



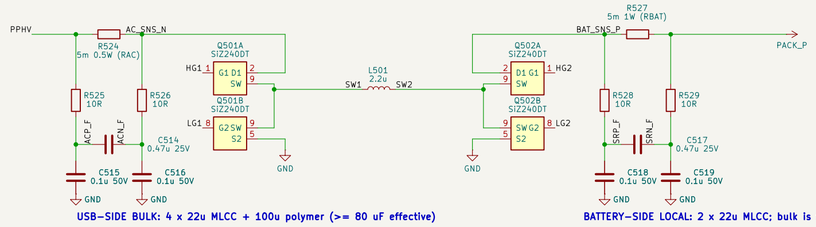
PP5V SUPPLY TPS62933 (1.2 MHz, 5.16 V, 3 x 22u out) — 5 V source + VCONN



PROBE PADS (1.0 mm): VBUS PPHV PP5V REGN SW1 SW2 + GND

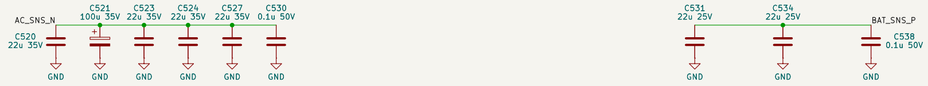


BQ25756 4-SWITCH BUCK-BOOST POWER STAGE (copy production layout exactly)

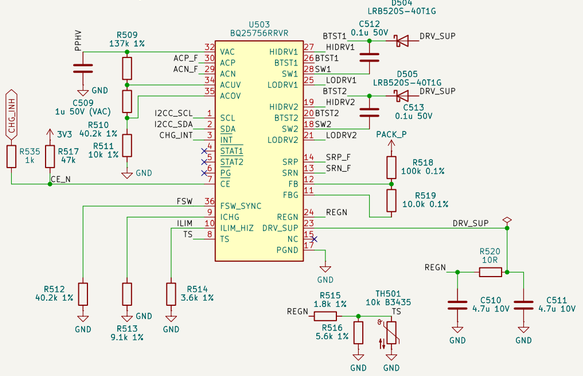


USB-SIDE BULK: 4 x 22u MLCC + 100u polymer (>= 80 uF effective)

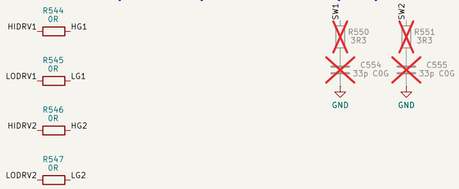
BATTERY-SIDE LOCAL: 2 x 22u MLCC; bulk is the shared PACK_P hybrid C208 (3V3 Rail sheet)



BQ25756 CONTROLLER (600 kHz, ICHG 5.5 A max, ILIM 5.6 A, VAC 4.1-22.5 V window)



GATE LINKS (0R, fitted) + SNUBBERS (DNP): PO tuning hedge



NOTES (this sheet is used twice: USB Port 1 = 4xx, USB Port 2 = 5xx)
1. Address straps are on the root sheet: port 1 ADCIN1 = LDO_3V3, ADCIN2 = GND (0x20); port 2 ADCIN1 = ADCIN2 = GND (0x21). Both are SafeMode (TPS25751 table 8-6).
2. Unused TPS25751 GPIOs (0, 1, 2, 4-7, 11) tied to GND per datasheet (KiCad ERC flags this; expected). Never configure them as outputs.
3. CC is protected by U405 (TPD2S300; CC short-to-VBUS OVP, 24 V DC, + IEC 61000-4-2 ESD; the TPS25751 CC pins are 26 V abs max but only +-1 kV HBM). VPWR/VM on LDO_3V3 (normal mode). VBIAS C451 0.1u 50V (>= 35 V needed). C401/C402 (220 pF) on the controller side: with the TPD2S300's 30-120 pF the total stays under 480 pF. D+ and D- are tied together at the connector (net DPDN_DCP; passive BC1.2 DCP); the TPS25751 does no BC1.2. No liquid detection and no SBU use on production (IP68 connectors, potted board): the connector SBU pins are no-connects; GPIO4-7 are tied to GND (note 2). U405 FLT -> GPIO3 = Fault_Input_Event (SBU_FLT, R441); a CC short to VBUS forces error recovery.
   D401 = TVS2200 flat clamp on VBUS (28 V at 40 A 8/20 us; TI's pick for the TPS25751).
4. Kelvin-route ACP/ACN from R424 and SRP/SRN from R427; filter parts at the IC.
5. Two MLCCs of each bulk bank plus C430/C438 at the FETs; the USB-side polymer C421 and the remaining MLCCs one row back. Polymer carries the bulk (no DC-bias loss), MLCCs carry the switching-frequency ripple. Battery side: 2 x 22u per port at the FETs: both ports, both TPS62933 inputs (1 x 10u 50V each) and the one shared PACK_P hybrid C208 form one bus: about 99 uF effective worst case, >= the BQ25756 80 uF C_OUT minimum each charger sees through its RBAT (Production/tools/pack_bus_sims.py A2). Still on hold: C427 (USB side; the USB sides are separate per port, so they can't share bulk).
6. I2Cc pull-ups go to this TPS25751's own LDO_3V3 (TI guidance).
7. The charger battery side is PACK_P directly; take SRN/FB from R427's pad, not from the pour.
8. VAC is taken before R424 per TI typical application; DRV_SUP fed from REGN via R420.
9. PPHV bulk (BQ25756 input bank) exceeds the 100 uF USB PD cSnkBulkPd: the TPS25751D sink switch soft-start limits inrush (C x SS: about 125 uF effective x 0.47 V/ms max = 0.06 A at SS setting 0). Set PPHV soft-start to setting 0 or 1 in the TPS25751 configuration.
10. Diamonds are PWR_FLAGs (ERC only, no part): each sits where its rail is sourced through a passive part.
11. CHG_INH: R417 pulls CE up to 3V3, so charging needs the MCU (DIO24) to drive CHG_INH low through R435. Reset, WDT reset, the ROM bootloader or an unprogrammed MCU stop charging: CC2340 Standby keeps the pin state. A dead pack (3V3 down) still charges. Keep DIS_CE_PIN = 0.
12. Gate drive: R441-R447 are 0R 0402 links at the gates (PO hedge, no dev board); swap to 2.2-10R if SW1/SW2 ring: R450/C454, R451/C455 are DNP 0402 RC snubbers (3R3 + 33p). No turn-off diodes; BTST straight to its cap. Keep the gate traces short and wide (TI: >= 0.5 mm) with the FETs right at the BQ25756. Each leg is one Vishay SiZ240DT half-bridge (Q401/Q402: unit A = high side, unit B = low side, pin 9 = SW). 40 V, 12.25/13.30 mOhm max at 4.5 V, Qg 6.9/6.5 nC. Worst case (20 V 5 A out from a 10 V pack) its fully-on high side dissipates ~1.8 W in one 3.3 x 3.3 package; needs the paddles' via arrays and the gap pad; the thermal throttle covers it. Buy from DigiKey/Mouser (JLC stock is near zero).
13. TP401 VBUS is a 1.0 mm probe pad. STAT1/STAT2/PG (U403 pins 4-6) are open-drain and unused (no pull-ups): status comes over I2C (TPS25751 I2Cr); set DIS_STAT_PINS = 1 in the BQ25756 configuration.
14. PP5V output: C441-C443 3 x 22u 1206 (about 33 uF effective) plus C407 10u at the PP5V pin. Each port's PP5V is its own TPS62933, a dedicated supply: USB PD cSrcBulk (10 uF min) applies, not the 120 uF cSrcBulkShared the TPS25751 table cites. Estimated 0.5 -> 3 A step undershoot about 190 mV (cap sims P3); measure it on PO.
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Spec + open items: Production/README.md
TPS25751D + BQ25756 + TPS62933; port 1 = 4xx, port 2 = 5xx
Monolith 100 W battery bank — production

Sheet: /USB Port 2/
File: usb_port.kicad_sch

Title: Monolith production board — USB-C port

Size: A2	Date: 2026-10-17	**Rev: P0**
KiCad E.D.A. 10.0.7		Id: 6/6